VIGIL-X4

Multi-domain robotics platform for reconnaissance, terrain exploration, and environmental threat sensing

The **VIGIL-X4** is a multi-domain robotics platform designed for tactical reconnaissance, hazardous-terrain exploration, and CBRN-related environmental threat sensing. Built around a rugged 15 × 20-inch, three-tier stepped chassis with all-terrain tires, the concept integrates three cooperating vehicle elements: a heavy-duty mothership rover, an aerial drone, and an autonomous micro-reconnaissance car.

Platform Architecture

The mothership uses a distributed dual-ESP32 architecture with ESP-NOW communication. The master controller coordinates the drive system, heading sensor, long-range telemetry, and remote-control interface. The slave controller manages the motorized garage mechanisms, GPS input, and environmental sensing suite.

Mobility, Navigation & Control

The master ESP32 is specified to manage dual 43A BTS7960 motor drivers, an HMC5883L digital compass, 433 MHz LoRa telemetry, and a 2.4 GHz FlySky remote-control link. Intended operating modes include manual control, autonomous operation, and garage/deployment mode. Actual autonomy and navigation performance should be documented through testing.

Sensing & Environmental Monitoring

The slave ESP32 is specified to interface with a NEO-6M GPS module and a sensor suite comprising an MQ-135 gas sensor, DHT11 temperature-and-humidity sensor, PIR motion sensor, flame sensor, rain sensor, and two ultrasonic sensors. These components provide environmental and proximity indications; they should not be represented as certified CBRN detection equipment without appropriate validated sensors and testing.

Cooperative Vehicle Elements

Mothership rover: The primary all-terrain platform carries the control electronics, sensors, telemetry equipment, and deployment bays.

Aerial drone: An elevated front flight deck is designed to dock a 15-inch carbon-fiber quadcopter for aerial observation.

Scout Mini Car: A rear garage bay with a motorized tailgate is designed to deploy a standalone micro-reconnaissance vehicle.

Robotic Manipulation & Surveillance

An open rear mezzanine deck carries a 5-DOF articulated robotic arm and an ESP32-CAM for close-up visual feedback and manipulation tasks. A front-mounted TP-Link Tapo IP camera provides the primary wide-area visual feed. These camera and arm functions should be described according to their implemented capabilities and verified demonstrations.

Mission Control Web HUD

A Mission Control web dashboard is intended to bring key telemetry and visual feeds into one interface. Suggested dashboard elements include connection status, rover mode, GPS position, heading, sensor readings, proximity alerts, battery status when available, and camera views. Only live, connected measurements should be shown as real-time data.

PROJECT NOTE • This document compiles the supplied project description into a structured overview. Hardware specifications, operational modes, and performance claims should be verified against the current build and test results before formal publication.